Harshith Bhattaram

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SUMMARY

Machine Learning Engineer with 3 years of experience building and deploying production ML systems across computer vision, predictive modeling, speech, and IoT. Skilled in model development, evaluation, MLOps, and scalable deployment using Python, PyTorch, XGBoost, MLflow, Kafka, Kubernetes & FastAPI. Currently focused on evaluating and optimizing speech ML architectures and data pipelines for Cantonese ASR.

EDUCATION

Master of Science, Computer Science
University at Buffalo, The State University of New York, USA

WORK EXPERIENCE

Vosyn Inc.

Machine Learning Engineer July 2026-Present

- Evaluated 6+ Cantonese speech corpora for audio, Hanzi, Jyutping coverage and labeling quality for architecture recommendation (direct audio-to-Jyutping) for the team's transcription pipeline.
- Benchmarked 3 Cantonese architectures (Wav2Vec2-BERT vs. whisper-turbo-pycantonese vs whisper-small-cantonese) across 2 datasets and 3 sample sizes, and 4 phonetic error metrics.
- Designed evaluation planning for 3 Cantonese speech to text pipelines using 2 models (whisper-turbo + pycantonese and alvanlii/whisper-small-cantonese) across 5 custom metrics with 4 sample sizes, catching sampling bias and hallucination patterns.

LocoNav Fleet Management Solutions

Software Engineer I (Machine Learning Engineer) Jun 2022-Jun 2024

- Deployed and optimized YOLOv8 and XGBoost drowsiness detection pipeline, exporting to ONNX and building TensorRT FP16 engines to reduce latency on edge devices.
- Built and containerized inference service with Docker and DeepStream SDK, implemented an event driven data pipeline using MQTT, Kafka, and FastAPI REST API with TimescaleDB to deliver real time alerts to the fleet dashboard team.
- Setup MLflow model registry and DVC based dataset versioning for staged model rollouts, orchestrated retraining workflows with Airflow and configuring EvidentlyAI for drift monitoring across production fleets.
- Integrated a LightGBM ETA model into a real time inference pipeline using Kafka telemetry and OSRM route features, reducing MAE to 10 minutes in production.
- Built and scaled the model serving layer using FastAPI, Redis, Kubernetes, with CI/CD pipelines via GitHub Actions and Docker, for deployment across 2 internal applications at sub 50ms latency across three regions.
- Implemented MLOps infrastructure using MLflow, Optuna and Airflow to operationalize the experiments into automated model retraining and hyperparameter tuning pipelines.

Software Engineer Intern (ML Intern) Jan 2022-May 2022

- Trained a LightGBM failure-prediction model on time-windowed IoT sensor features (rolling means, slopes, fault-code frequency) with Isolation Forest anomaly scores, across 12,000+ fleet vehicles.
- Designed a time-aware and leakage-safe feature pipeline on Kafka and Spark infrastructure, registered in Feast for consistent training and feature computation.
- Containerized and shipped the scoring model as a FastAPI service on Kubernetes with canary rollouts via MLflow's model registry, catching firmware-driven feature drift issues during shadow deployment before it reached production.

PROJECTS

Machine Learning Research Assistant (Dept of CSE, University at Buffalo) March 2026-Present

- Built multimodal RAG pipeline on MIMIC-CXR dataset using BioMedCLIP ViT-B/16 embeddings, FAISS for retrieval and BM25 hybrid search with Reciprocal Rank Fusion.
- Orchestrated retrieval and generation with LangChain as a sequence of callable tools covering image embedding, FAISS search, zero-shot BioMedCLIP captioning and radiology report synthesis with Groq Llama 4 Scout.
- Evaluated pipeline quality using BERT Score, BLEU, and ROUGE across 27K filtered test samples through a custom evaluation harness that validated FAISS index integrity, deduplication in retrieval, and cross-stage data alignment.

SKILLS

- **Machine Learning, Deep Learning & Evaluation:** Python, PyTorch, Tensorflow, Scikit-learn, Numpy, Pandas, XGBoost, Yolo v8, Wav2Vec2m Whisper, BiomedCLIP, LSTM, Time-Series Analysis, Feature Engineering, Speech/ASR Evaluation
- **Generative AI & NLP:** HuggingFace, Transformers, PEFT/LoRA/QLoRA, RAG, FAISS, BM25, RRF, LangChain, LLM APIs, Prompt Engineering, LLM Fine-Tuning, vLLM, LiteLLM
- **MLOps, Deployment & Data Infrastructure:** FastAPI, Docker, Kubernetes, Git, MLflow, DVC, Airflow, Kafka, Spark, Redis, Feast, PostgreSQL, TimescaleDB, TesnroRT, ONNX, NVIDIA DeepStream, Evidently AI, GCP